

Submission LINALG1-PS1-SUB08

Question LINALG1-PS1-Q01

scan faint / margin cut off

visible work:

$$\det = 6(5) - 1(4) = 26 \dots ?$$

last line is hard to read; maybe $x=1$, $y=2$

Question LINALG1-PS1-Q02

scan faint / margin cut off

visible work:

$$\det(A) = 2 \cdot 3 - 2 \cdot 2 = 2 \dots ?$$

last line is hard to read; maybe $\det=2$; invertible

Question LINALG1-PS1-Q03

scan faint / margin cut off

visible work:

$$T(1, -1) = (1 - 2, 3 + 1) = (-1, 4) \dots ?$$

last line is hard to read; maybe $T(1, -1) = (-1, 4)$; matrix $\begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}$; linear

Question LINALG1-PS1-Q04

scan faint / margin cut off

visible work:

$$\text{Set } c \blacksquare(1, 0) + c \blacksquare(0, 1) = (0, 0) \dots ?$$

last line is hard to read; maybe only trivial combination; $\{(1, 0), (0, 1)\}$ is linearly independent

Question LINALG1-PS1-Q05

scan faint / margin cut off

visible work:

$\text{Null}(A)$ is the set of vectors sent to the zero vector.... ?

last line is hard to read; maybe the vectors mapped to zero; exactly the solution space of $Ax=0$

Question LINALG1-PS1-Q06

scan faint / margin cut off

visible work:

$$\det = 5(3) - 2(3) = 9 \dots ?$$

last line is hard to read; maybe $x=1$, $y=2$